SIMATS ENGINEERING
ASSESSMENT TOOL 2

COURSE NAME: COMPUTER NETWORKS
COURSE CODE: CSA07

COURSE OUTCOME COVERED

CO2: Configure IP addressing schemes, subnetting, and basic routing techniques using simulation tool, for efficient network design and topology. *(BL3)*

Assessment Tool Weightage: 30%

Code of Conduct:

I B Sreesha 192126O034 *(Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.*
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: Sreesha

Criteria	Weightage (%)	Q1 10marks	Q2 10marks	Q3 10marks	Q4 10marks
Understanding of Scenario	20%	2	1	1	1
Identification of Key Issues	20%	1	1	1	1
Application of Concepts	25%	1	1	1	1
Decision Making	20%	1	1	1	1
Clarity of Response	15%	1	1	1	1

Q5 [illegible] mark
2
1
1
1
1

8

26/5

Annexure A

Scenario-Based

1.A newly established Smart Campus is deploying a secure network infrastructure for its academic and administrative departments. The network administrator has been allocated the IP address block 192.168.100.0/24. Based on the expected number of devices, the following subnet masks have already been assigned:

School of Computing → /25 School of Electronics → /26

School of Mechanical Engineering → /27

As the network engineer, design the subnet allocation by performing the following tasks:

1. Determine the network address and broadcast address for each department.
2. Identify the valid host IP address range for each subnet.
3. Calculate the number of usable host addresses available in each subnet.
4. Determine the remaining unused IP address range within the given network block.

2.A multinational software company is expanding its headquarters and has been assigned the private IP block **10.10.0.0/16**. Different departments require different subnet sizes based on the number of employees. The network administrator has allocated the following subnet masks:

Software Development → **/24**
Quality Assurance → **/25** Human Resources → **/26** Executive Management → **/27**

As the network administrator, perform the following tasks:

1. Determine the **network address** and **broadcast address** for each department.
2. Identify the **valid range of host IP addresses**.
3. Calculate the **number of usable hosts** in each subnet.
4. Identify the **unused IP address range** remaining within the allocated IP block.

3. A smart hospital is migrating its network infrastructure from IPv4 to IPv6 to support thousands of connected medical devices and IoT sensors. The hospital has been assigned the IPv6 network:

   2001:0db8:abcd:0000:0000:0000:0000:0000/64

   As the network engineer responsible for the migration, perform the following:

1. Convert the given IPv6 address into its **compressed notation**.
2. Expand the compressed address back into its **full hexadecimal representation**.
3. Identify the **network prefix** using the **/64** prefix length.
4. Calculate the **total number of host addresses** supported within the subnet.
5. the total number of possible host addresses available within this network.

4. An Internet Service Provider (ISP) connects multiple customer networks through a core router. The router maintains the following routing table entries:

   192.168.1.0/24

   192.168.2.0/24

   192.168.3.0/24

   A packet arrives at the router with the destination IP address:

   192.168.2.45

   As the routing engineer, determine:

1. The **destination subnet** to which the packet belongs.
2. The **matching routing table entry**.

4. The **default route** that should be used if the destination does not match any routing table entry.

5.A company establishes two office departments connected through a Layer-3 router. The allocated subnet information is as follows:

Administration LAN → **192.168.10.0/26**

Finance LAN → **192.168.10.64/26**

The router interfaces are configured with:

Administration Gateway → **192.168.10.1**

Finance Gateway → **192.168.10.65**

As the network administrator, perform the following tasks:

1. Determine the **network address** and **broadcast address** for each LAN.
2. Identify the **valid host IP address range**.
3. Suggest suitable IP addresses for **three computers** in each LAN.
4. Identify the **default gateway** that hosts in each LAN should use for communication.

Question 1 – Smart Campus IPv4 Subnet Allocation

1. Understanding the scenario

The Smart Campus has been allocated the IPv4 address block 192.168.100.0/24. Three departments require different subnet sizes:

- School of Computing: `/25`
- School of Electronics: `/26`
- School of Mechanical Engineering: `/27`

The objective is to allocate non-overlapping subnets, calculate network and broadcast addresses, identify valid host ranges, and determine the unused address range.

1. Subnet allocation and calculations

The original `/24` network contains 2^32-24=256total IP addresses.

Department	Subnet mask	Total addresses	Usable hosts
School of Computing	/25 (255.255.255.128)	128	126
School of Electronics	/26 (255.255.255.192)	64	62
School of Mechanical Engineering	/27 (255.255.255.224)	32	30

Usable IPv4 host addresses are calculated using:

\[\text{Usable hosts}=2^{(32-\text{prefix})}-2 \]

Two addresses are excluded: the network address and the broadcast address.

1.3 Detailed subnet allocation

The subnets are allocated sequentially, starting from the lowest available address. Each subnet begins at a valid boundary for its prefix length.

A. School of Computing – 192.168.100.0/25

•Network address: `192.168.100.0`
•Broadcast address: `192.168.100.127`
•Valid host range: `192.168.100.1 – 192.168.100.126`
•Usable host addresses: 2^7-2=126

B. School of Electronics – 192.168.100.128/26

•Network address: `192.168.100.128`
•Broadcast address: `192.168.100.191`
•Valid host range: `192.168.100.129 – 192.168.100.190`
•Usable host addresses: (2^6-2=62)

C. School of Mechanical Engineering – 192.168.100.192/27

•Network address: `192.168.100.192`
•Broadcast address: `192.168.100.223`
•Valid host range: `192.168.100.193 – 192.168.100.222`
•Usable host addresses: 2^5-2=30

- ### Remaining unused IP address range

The three subnets consume:

128+64+32=224

The original `/24` block contains 256 addresses, so:

256-224=32

The unused range is 192.168.100.224 – 192.168.100.255, which corresponds to `192.168.100.224/27`.

- ### Decision and conclusion

The proposed allocation uses 224 of the 256 addresses without overlap and leaves 32 addresses available for future campus expansion. The department with the largest expected device count receives the largest subnet, while smaller departments receive smaller subnets.

Final result: Computing receives 126 usable hosts, Electronics receives 62, Mechanical Engineering receives 30, and 32 addresses remain unused.

Question 2 – Multinational Software Company IPv4 Subnetting

2.1 Understanding the scenario

The company has received the private IPv4 address block 10.10.0.0/16. Four departments require different subnet sizes according to their employee counts:

- Software Development: `/24`
- Quality Assurance: `/25`
- Human Resources: `/26`
- Executive Management: `/27`

The goal is to assign non-overlapping subnets while preserving as much address space as possible for future departments and devices.

2.2 Subnet size calculations

Department	Prefix	Subnet mask	Total addresses	Usable hosts
Software Development	/24	255.255.255.0	256	254
Quality Assurance	/25	255.255.255.128	128	126
Human Resources	/26	255.255.255.192	64	62
Executive Management	/27	255.255.255.224	32	30

The usable host formula is:

Usable hosts=2^(32-prefix)-2

2.3 Detailed subnet allocation

Assuming sequential allocation from the beginning of the `/16` block:

Department	Network address	Broadcast address	Valid host range
Software Development	`10.10.0.0/24`	`10.10.0.255`	`10.10.0.1 – 10.10.0.254`
Quality Assurance	`10.10.1.0/25`	`10.10.1.127`	`10.10.1.1 – 10.10.1.126`
Human Resources	`10.10.1.128/26`	`10.10.1.191`	`10.10.1.129 – 10.10.1.190`
Executive Management	`10.10.1.192/27`	`10.10.1.223`	`10.10.1.193 – 10.10.1.222`

The allocation is valid because each network starts on the appropriate subnet boundary, and none of the address ranges overlap.

2.4 Remaining unused IP address range

The original `/16` block contains: 2^(32-16)=65,536 total addresses The allocated subnets

consume: 256+128+64+32=480 text addresses Therefore:

65,536-480=65,056unused addresses

The remaining unused address range, under the stated sequential allocation, is:

`10.10.1.224 – 10.10.255.255`

This range includes 65,056 addresses. The range starts immediately after the last allocated subnet and continues to the end of the `/16` block.

2.5 Decision and conclusion

The `/24` subnet is assigned to Software Development because it receives the largest address pool. The other departments receive progressively smaller subnets based on their requirements.

This allocation conserves address space, avoids overlapping networks, simplifies routing, and preserves a large pool for future expansion.

Final result: The four departments receive 254, 126, 62, and 30 usable host addresses, respectively, while 65,056 addresses remain unused.

Question 3 – Smart Hospital IPv6 Addressing

1. Understanding the scenario

The smart hospital is migrating from IPv4 to IPv6 to support a large number of connected medical devices and IoT sensors. The assigned IPv6 network is:

`2001:0db8:abcd:0000:0000:0000:0000:0000/64`

The tasks involve compressing and expanding the IPv6 address, identifying the `/64` network prefix, and calculating the total number of available addresses.

- ### Convert the address into compressed notation

IPv6 addresses consist of eight hexadecimal groups, called hextets, separated by colons. Leading zeros within a hextet may be removed, and one consecutive sequence of zero-valued hextets may be replaced with `::`.

Given address:

`2001:0db8:abcd:0000:0000:0000:0000:0000/64`

After removing leading zeros and compressing the consecutive zero groups, the result is:

Compressed IPv6 address: `2001:db8:abcd::/64`

The `::` represents the consecutive omitted zero-valued hextets.

3. Expand the compressed address

To expand the address, replace `::` with enough zero-valued hextets to obtain exactly eight groups.

Full expanded IPv6 address:

`2001:0db8:abcd:0000:0000:0000:0000:0000/64`

This is the same address as the original network.

- ## Identify the network prefix

The prefix length `/64` means that the first 64 bits identify the network prefix, while the remaining 64 bits form the interface identifier portion.

Component	Value
IPv6 network prefix	`2001:0db8:abcd:0000::/64`
Network bits	64 bits
Remaining interface identifier bits	64 bits
Total address length	128 bits

The first four hextets, `2001:0db8:abcd:0000`, form the 64-bit network prefix.

3.5 Calculate the total number of addresses

IPv6 has 128 bits. For a `/64` subnet, the remaining address bits are: 128-64=64 bits

The total number of addresses is: 2^64=18,446,744,073,709,551,616

Therefore, the subnet contains 18,446,744,073,709,551,616 possible IPv6 addresses.

Unlike conventional IPv4 subnet calculations, IPv6 does not use a broadcast address, and the IPv4 rule of subtracting two addresses should not be applied here. The value above is the total size of the `/64` address space.

3.6 Decision and conclusion

The `/64` prefix provides an enormous address space suitable for a smart hospital with medical equipment, monitoring systems, sensors, and other connected devices. IPv6 addressing also removes the need for conventional IPv4 broadcast addressing.

Final result: The compressed address is `2001:db8:abcd::/64`, the expanded address is `2001:0db8:abcd:0000:0000:0000:0000:0000/64`, and the subnet contains \(2^{64}\) possible addresses.

Question 4 – ISP Routing Table and Packet Forwarding

- ### Understanding the scenario

An Internet Service Provider (ISP) uses a core router to connect multiple customer networks. The router has the following routing table entries:

- `192.168.1.0/24`
- `192.168.2.0/24`
- `192.168.3.0/24`

A packet arrives with the destination IP address 192.168.2.45. The router must identify the matching subnet, select the appropriate forwarding route, and determine what to do if no route matches.

- ### Identify the destination subnet

The destination IP address is:

`192.168.2.45`

The subnet mask for a `/24` network is `255.255.255.0`. Applying the subnet mask to the

destination address gives: `192.168.2.45` AND `255.255.255.0` = `192.168.2.0`

Therefore, the destination belongs to the network:

`192.168.2.0/24`

- ### Identify the matching routing table entry

Routing table entry	Match?	Reason
`192.168.1.0/24`	No	Different third octet

192.168.2.0/24	**Yes**	**Matches the destination network**
192.168.3.0/24	No	Different third octet

The matching route is 192.168.2.0/24.

The router uses the destination IP address to search its routing table and selects the matching route. When multiple routes match, it generally selects the route with the longest matching prefix.

4.4 Determine the next-hop network/interface

The router forwards the packet through the outgoing interface connected to the 1. /24 destination network, if that network is directly connected.

If the network is reached through another router, the packet is sent to the next-hop IP address specified in the routing table.

The scenario does not provide the router's interface names, next-hop IP address, or whether the network is directly connected. Therefore, the exact interface or next-hop address cannot be determined from the given information alone.

The logical forwarding decision is:

- Destination network: 192.168.2.0/24
- Destination host: 192.168.2.45
- Forwarding decision: Use the route for 192.168.2.0/24

• Determine the default route

If the destination does not match any specific routing table entry, the router can use a default route, provided one is configured.

The default route is:

0.0.0.0/0

This route matches any IPv4 destination address. Because its prefix length is zero, it is less specific than the three listed /24 routes and is used only when no more-specific matching route is available.

If no default route exists, the router generally drops the packet and may return an ICMP Destination Unreachable message.

• Decision and conclusion

The router should forward the packet for 192.168.2.45 using the route 192.168.2.0/24. This is the most specific matching entry among the provided routes. A configured default route, 0.0.0.0/0, provides a fallback for destinations that do not match any specific route.

Final result: Destination subnet — `192.168.2.0/24`; matching route — `192.168.2.0/24`; forwarding interface — the interface or next hop associated with that route; default route — `0.0.0.0/0`, if configured

Question 5 – Administration and Finance LAN Configuration

5.1 Understanding the scenario

A company has two office departments connected through a Layer-3 router. Each department has its own IPv4 subnet and default gateway.

The given configuration is:

Department	Network address	Configured gateway
Administration LAN	`192.168.10.0/26`	`192.168.10.1`
Finance LAN	`192.168.10.64/26`	`192.168.10.65`

The objective is to determine the network and broadcast addresses, valid host ranges, suitable addresses for three computers in each LAN, and the appropriate default gateway.

2. Calculate the subnet boundaries

Both LANs use the `/26` subnet mask:

`255.255.255.192`

The number of addresses per subnet is:

2^(32-26)=2^6=64

The usable host count is:

64-2=62

Each subnet therefore contains 64 total addresses and 62 usable host addresses.

- Network address, broadcast address, and host range

Property	Administration LAN	Finance LAN
Network address	`192.168.10.0`	`192.168.10.64`
Subnet mask	`255.255.255.192`	`255.255.255.192`
Broadcast address	`192.168.10.63`	`192.168.10.127`

Valid host range	**192.168.10.1 – 192.168.10.62**	**192.168.10.65 – 192.168.10.126**
Usable hosts	62	62

The two subnets do not overlap. Administration uses addresses from `.0` to `.63`, while Finance uses addresses from `.64` to `.127`.

5.4 Suggest suitable IP addresses for three computers in each LAN

The router gateway addresses are already assigned, so the computer addresses must be different from the gateways and from one another.

Administration LAN

Device	IP address	Default gateway
Computer A1	192.168.10.2	192.168.10.1
Computer A2	192.168.10.3	192.168.10.1
Computer A3	192.168.10.4	192.168.10.1

All three computer addresses are within the valid Administration host range.

Finance LAN

Device	IP address	Default gateway
Computer F1	192.168.10.66	192.168.10.65
Computer F2	192.168.10.67	192.168.10.65
Computer F3	192.168.10.68	192.168.10.65

All three computer addresses are within the valid Finance host range.

These are suitable example addresses, assuming they are not already in use or reserved by another device. Static assignments or DHCP reservations can be used to prevent duplicate IP addresses.

5. Identify the default gateway for each LAN

The default gateway is the router interface that a host uses to reach destinations outside its own subnet.

- Administration LAN: `192.168.10.1`
5. Finance LAN: `192.168.10.65`

For example, if a computer with address `192.168.10.2` communicates with another computer in the Administration subnet, it can communicate directly over the LAN. If it needs to reach a host in the Finance subnet, it sends the packet to `192.168.10.1`, assuming the routing configuration is correct.

Similarly, a Finance computer with address `192.168.10.66` sends packets destined for the Administration LAN to `192.168.10.65`.

The Layer-3 router must have both LAN interfaces configured with the specified gateway addresses and must have the appropriate connected routes enabled.

5.6 Decision and conclusion

The `/26` subnet mask provides 62 usable host addresses for each department. The configured gateways are valid host addresses within their respective subnets. The suggested computer addresses are unique, fall within the correct host ranges, and leave room for additional devices.

Final result: Administration uses `192.168.10.0/26` with gateway `192.168.10.1`, and Finance uses `192.168.10.64/26` with gateway `192.168.10.65`. The Layer-3 router enables communication between the two LANs when its interfaces and routing are configured correctly.

RUBRICS FOR CALCULATION

Scenario Based Assessment					
Criteria	**Weightage (%)**	**Level 4 (Exemplary)**	**Level 3 (Proficient)**	**Level 2 (Developing)**	**Level 1 (Beginning)**
Understanding of Scenario	20%	Demonstrates clear and complete understandin g of the scenario and context	Shows good understandi ng with minor gaps	Partial understandin g; misses key aspects	Misinterpret s or fails to understand the scenario
Identification of Key Issues	20%	Accurately identifies all relevant issues and factors involved	Identifies most key issues with minor omissions	Identifies limited issues; some irrelevant points	Unable to identify key issues
Application of Concepts	25%	Applies appropriate concepts effectively to address the scenario	Applies concepts with minor errors	Limited or partially correct application	Unable to apply relevant concepts
Decision Making	20%	Makes well-reasoned, logical decisions	Makes reasonable decisions	Makes weak decisions with limited justification	No clear decisions made or

		with strong rationale	with some justification		rationale provided
Clarity of Response	15%	Response is clear, structured, and logically presented	Generally clear with minor issues	Somewhat unclear or poorly structured	Disorganized and difficult to understand